



Zamson Lim Zhen Pin

SOC ENGINEERING · DETECTION · PURPLE TEAM

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Final-year BSc (Hons) Cybersecurity student at Asia Pacific University. After a ransomware incident left my employer unable to reconstruct what had happened, I proposed a Security Operations Centre, got approval, and **built and ran it as the only person on security** — architecture, detection engineering, incident response and reporting.

EXPERIENCE

Network & Security Administrator (Intern)

Aug 2025 – Aug 2026

Palmgold Management Sdn Bhd · Kuala Lumpur · sole owner of security operations

- **Built the company's Security Operations Centre from nothing** after an August 2025 ransomware incident, taking it from a spare-desktop proof of concept to the production stack that now runs the company's monitoring.
- **Deployed a Wazuh SIEM with co-located OpenSearch** on a Proxmox host I set up myself, integrating firewall, core and access switching, wireless, Windows endpoint, Active Directory and Zeek telemetry through custom ingest pipelines.
- **Rolled out monitoring agents to 80+ Windows machines** via Active Directory Group Policy, using an idempotent installer that was safe to re-run and logged per machine.
- **Built the network sensor** — switch port mirroring into a multi-node Zeek cluster through a Python decapsulation stage — giving visibility into internal traffic the firewall never sees.
- **Wrote the detection content:** custom decoders and correlation rules matched to the hardware actually in use, deliberately without an endpoint-agent dependency I had no authority to deploy fleet-wide.
- **Ran authorized purple-team exercises**, attacking from Kali Purple then detecting from the SIEM, and turned each into detection content; surfaced an unmonitored network segment that no rule could have covered.
- **Ran an authorized internal vulnerability assessment** and delivered three audience-tailored reports, honestly downgrading one finding to "unconfirmed" when it could not be reproduced.
- **Recovered a full ingestion outage** in which archive logging filled the disk and silently flipped every index read-only, then installed retention policies so it could not recur.
- **Built the SOC's operator dashboards**, served from the SIEM host behind a hardened reverse proxy, and handed over a documented, running SOC at the end of the internship.

SKILLS

SIEM	Wazuh (deployment, agent management, custom decoders and correlation rules), OpenSearch query DSL and aggregations, ingest pipeline design, file integrity monitoring, alert triage and incident response
NETWORK	Zeek network security monitoring, FortiGate policy and syslog integration, Aruba switching and wireless, traffic mirroring and decapsulation, Wireshark, Cisco Packet Tracer
SYSTEMS	Proxmox virtualization, Ubuntu / Rocky Linux administration, systemd tuning, Nginx reverse proxy and TLS, Windows Server, Active Directory and Group Policy at fleet scale
SECURITY	Purple-team exercise design and execution, authorized vulnerability assessment (OpenVAS, Nmap, Nikto), malicious code analysis in isolated labs, Autopsy forensics, Kali Linux
CODE	Python, JavaScript (D3.js, Three.js), Bash, Windows batch, Java, C++, C#, Rust
LANGUAGES	English, Mandarin, Malay, Cantonese

PROJECTS

Purple-team exercise: tunnelled C2

- Set up the attack myself — a remote-control server reached through a tunnel so its real address never showed — then switched sides and hunted it from the SIEM.
- The firewall and the tunnel both showed only a shared address; the DNS records were the only thing that named the infected machines.
- Tested honestly: a plain implant to an ordinary server was **missed**, because the tunnel was itself the signature.

Zeek network monitoring pipeline

- Mirrored traffic decapsulated in Python and replayed onto a virtual interface into a multi-node Zeek cluster.
- Detections for domain-generation bursts, DNS tunnelling, exfiltration and inbound scans.
- Re-characterised an alarming reported capture loss as an MTU artefact of the mirror tunnel — which decides which detections can be trusted.

Ransomware canary detection

- File integrity monitoring rules scoped to a canary folder, mapped to MITRE T1486, to catch mass-encryption behaviour early.
- Validated with a benign encryptor hard-guarded to that folder — never a live sample.
- Path-scoping was the fix after fleet-wide scoping fired on every agent's overnight scan and spammed the whole estate.

SOC operator dashboards

- Three single-file consoles built for daily use: a D3 world traffic map, an endpoint alert console, and a Three.js traffic visualizer.
- De-identified and published as open-source templates; live demos and source linked from the portfolio.

Wireless & access point monitoring

- Wireless syslog brought into the SIEM with custom decoders for association, authentication and roaming events.
- A scheduled enrichment pipeline resolving hardware addresses to hostnames, so an alert names a machine rather than a MAC.

Malicious file analysis (Autopsy)

- Forensics exercise at APU examining a lecturer-provided sample in Autopsy.
- Worked through the file's contents and metadata rather than trusting a scanner verdict; written up as a formal report.

Active Directory monitoring

- Learned AD properly — domain structure, Group Policy, event log semantics — then built monitoring on native Windows event logs.
- Saved searches for authentication failure, log clearing and account enumeration, ranked by volume so an unusual spike surfaces early.

Authorized vulnerability assessment

- Internal assessment from Kali Purple using OpenVAS, with Nmap, Nikto and command-line work to verify by hand what the scanner claimed.
- Findings included unauthenticated printer administration, exposed print services and a reverse-proxy bypass on an office device.
- Delivered as three reports for three audiences; re-scanned and downgraded one finding to "unconfirmed" when it would not reproduce.

Alerting pipeline

- Custom Python integration pushing critical alerts to chat, so triage never depended on somebody watching a dashboard.
- De-duplicated with file locking and checked against the API response, and tolerant of binary content that would otherwise break the send.

Building with AI tooling

- Built and debugged much of the SOC tooling with Claude Code, and ran local models through Ollama to judge what genuinely needs a hosted one.
- The discipline is the point: what to delegate, what to verify, and never trusting a "done" message over the artefact itself.

EDUCATION & COMPETITIONS

BSc (Hons) Cybersecurity

2024 – 2027

Asia Pacific University · Kuala Lumpur · final year

- Coursework across network security, digital forensics and secure development.

Capture The Flag

2024 – present

SekaiCTF · CYNX · BeSec · picoCTF

- SekaiCTF 2025 with team Digital Phantoms: solved a medium reverse-engineering challenge on an Android APK using jadx-gui, keytool and Python.